Definition

If a line touches the given circle at only one point, then it is called tangent to the circle.

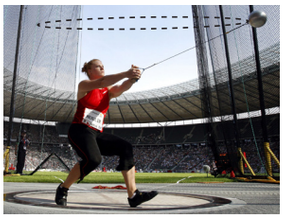
Real life examples of tangents to circles

(i) When a cycle moves along a road, then the road becomes the tangent at each point when the wheels rolls on it.



Fig. 4.53(a)

(ii) When a stone is tied at one end of a string and is rotated from the other end, then the stone will describe a circle. If we suddenly stop the motion, the stone will go in a direction tangential to the circular motion.



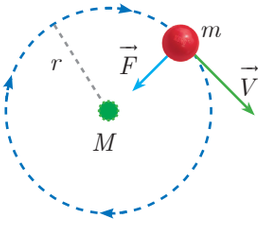


Fig. 4.53(b)

Some results on circles and tangents

1. A tangent at any point on a circle and the radius through the point are perpendicular to each other.

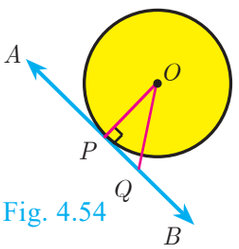


Fig. 4.54

2. (a) No tangent can be drawn from an interior point of the circle.

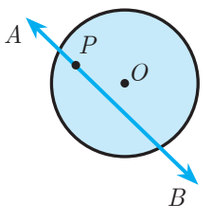


Fig. 4.55(a)

(b) Only one tangent can be drawn at any point on a circle.

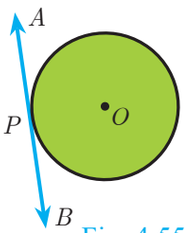


Fig. 4.55(b)

(c) Two tangents can be drawn from any exterior point of a circle.

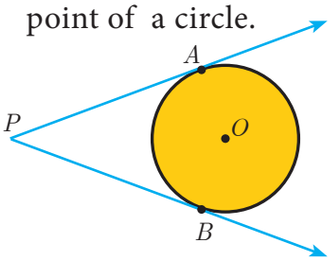


Fig. 4.55(c)

3. The lengths of the two tangents drawn from an exterior point to a circle are equal,

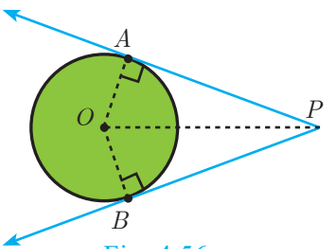


Fig. 4.56

Proof : By 1. $OA \perp PA, OB \perp PB$. Also $OA = OB =$ radius, OP is common side. $\angle AOP = \angle BOP$

Therefore, by SAS Rule $\Delta OAP \cong \Delta OBP$. Hence $PA = PB$

4. If two circles touch externally the distance between their centers is equal to the sum of their radii, that is $OP = r_1 + r_2$

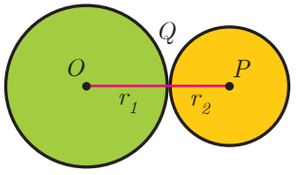


Fig. 4.57

Proof : Let two circles with centers at O and P touch other at Q.

Let $OQ = r_1$ and $PQ = r_2$ and let $r_1 > r_2$.

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